



FISHLINIC ·
CAPSTONE 2026

Smart Aquaculture

*powered by **AI** · **Computer Vision** · **IoT***

An autonomous tank that monitors itself, diagnoses fish health, controls hardware, and explains everything in plain English.

9

MODULES

5

AI MODELS

3

CLIENTS

60s

SCHEDULER TICK

THE PROBLEM

Aquaculture is a \$300B industry — still run with thermometers and gut feel.



Manual water testing

Hobbyists test pH, DO₂, temperature 1–2× per week with strips. Fish die from chemistry shifts that happen in hours.



Disease caught too late

Visible symptoms mean disease has spread for days. Without continuous monitoring there is no early warning.

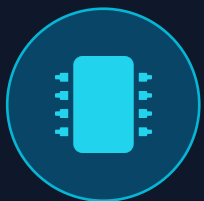


No automation

Feeders run on dumb timers. Lights ignore sensors. Pump on/off is manual. The system can't react to its own state.

→ We built the system that does all three. Continuously. Autonomously. Explainably.

Fishlinic — one connected stack from sensor to AI



SENSE

Arduino sensors stream pH, temperature, dissolved O₂, CO₂ at 1 Hz over USB serial. Mock mode keeps demos working without hardware.



REASON

Veronica — local LLM agent (Ollama + qwen2.5:3b) with tool-calling. Reads sensors, controls hardware, explains decisions.



SEE

YOLOv11 disease detector + counting model + ConvLSTM-VAE behavior anomaly detector run continuously on the camera feed.

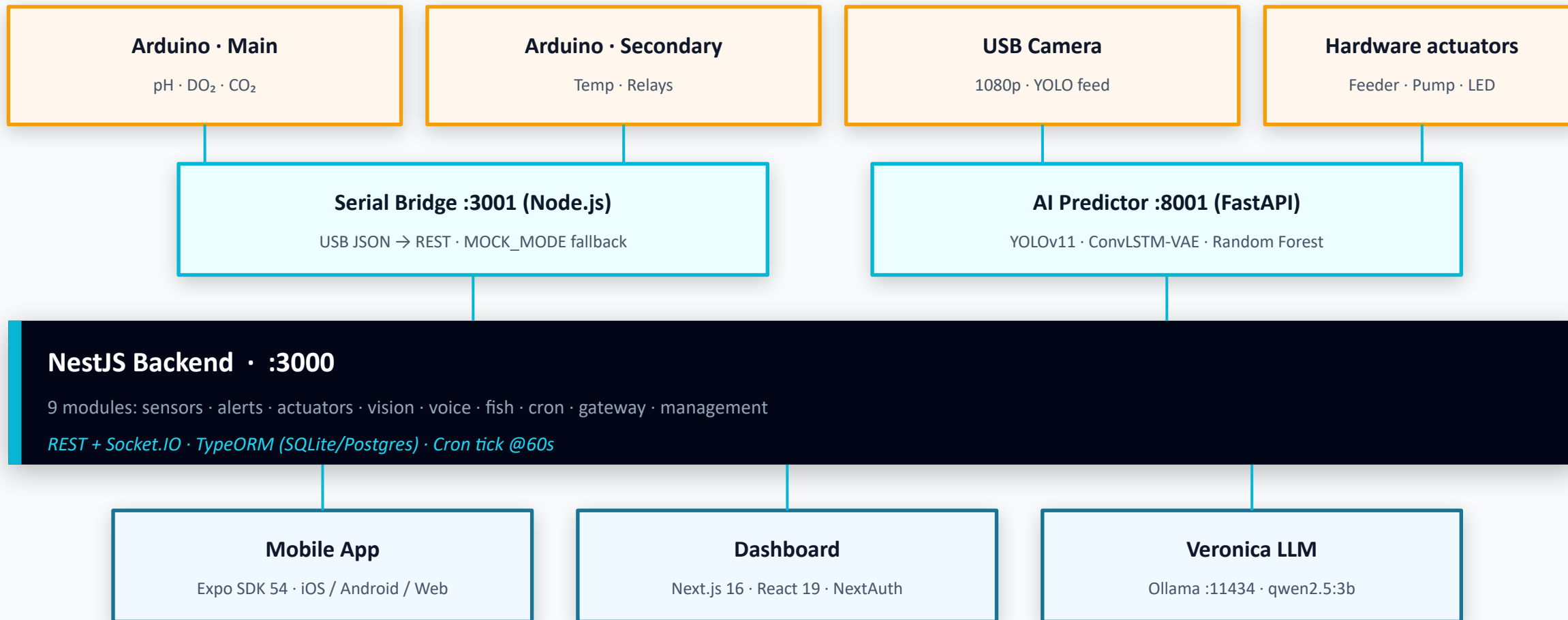


ACT

Relay-driven feeder, air pump, LED strip. Cron-style scheduler orchestrates feed times, lighting cycles, and emergency overrides.

All four pillars feed each other. Reason → Act loop closes the system.

Five services. One source of truth.



Production-grade tooling, end to end.



Backend

NestJS · TypeScript
TypeORM · Socket.IO
Class-validator



Mobile

Expo SDK 54 · RN 0.81
React 19 · React Native
Keyboard Controller



Dashboard

Next.js 16 · React 19
Tailwind · NextAuth
Server Components



AI Pipeline

FastAPI · PyTorch
Ultralytics · scikit-learn
OpenCV



LLM Agent

Ollama · qwen2.5:3b
Tool-calling protocol
Local inference



Storage

PostgreSQL (prod)
SQLite (dev)
better-sqlite3 driver

Meet *Veronica* —

the AI agent that actually runs the tank.



Local LLM

Runs entirely on-device via Ollama. No cloud, no API fees, no privacy concerns.



Tool-calling

Reads sensors, controls hardware, queries history. 6 tools wired into the agent loop.



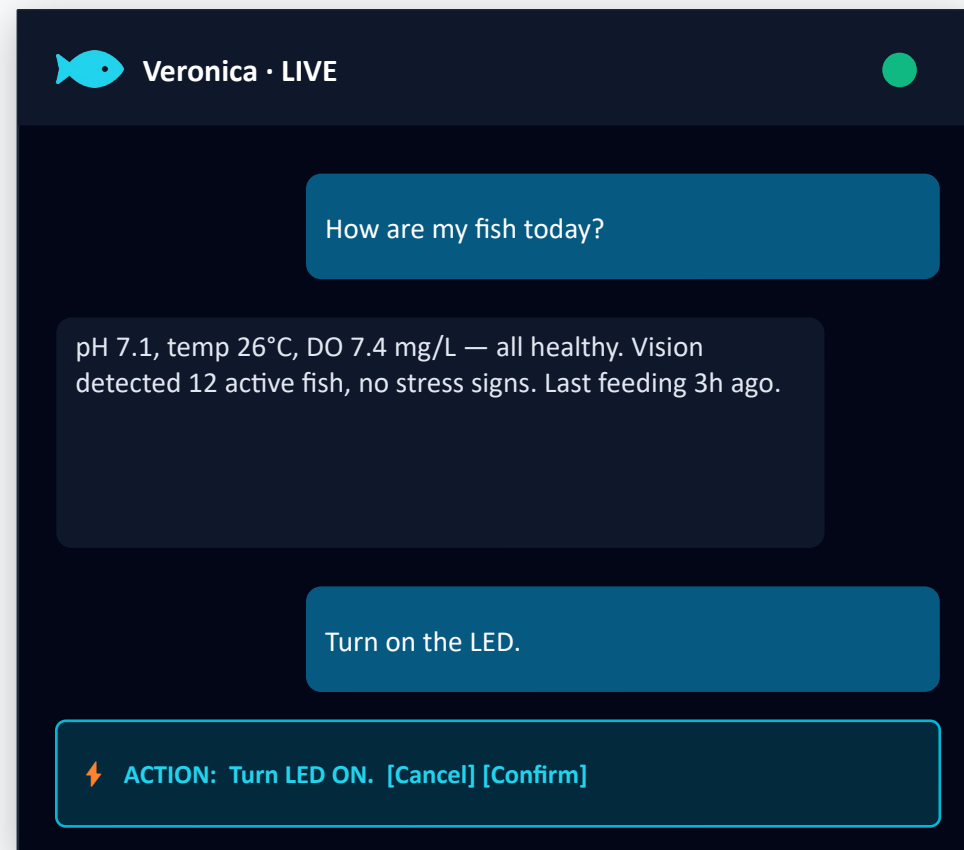
Confirm-before-act

User mode requires confirmation. Auto mode lets Veronica respond to emergencies alone.



Sessioned chat

Persists every conversation. Restores context across app reopens.



Three models. One camera feed.

YOLOv11 · DISEASE

yolo_disease.pt

Trained on a curated set of fungal infection, fin rot, ich, and ulcer images. Runs every camera tick when fish are visible. Bounding boxes + confidence per disease class.

8 classes

PATHOLOGIES

YOLOv11 · COUNT

yolo_count.pt

Counts active fish per frame. Drives the population badge in the UI and detects population drops that may indicate mortality events.

± 1 fish

ACCURACY

ConvLSTM-VAE

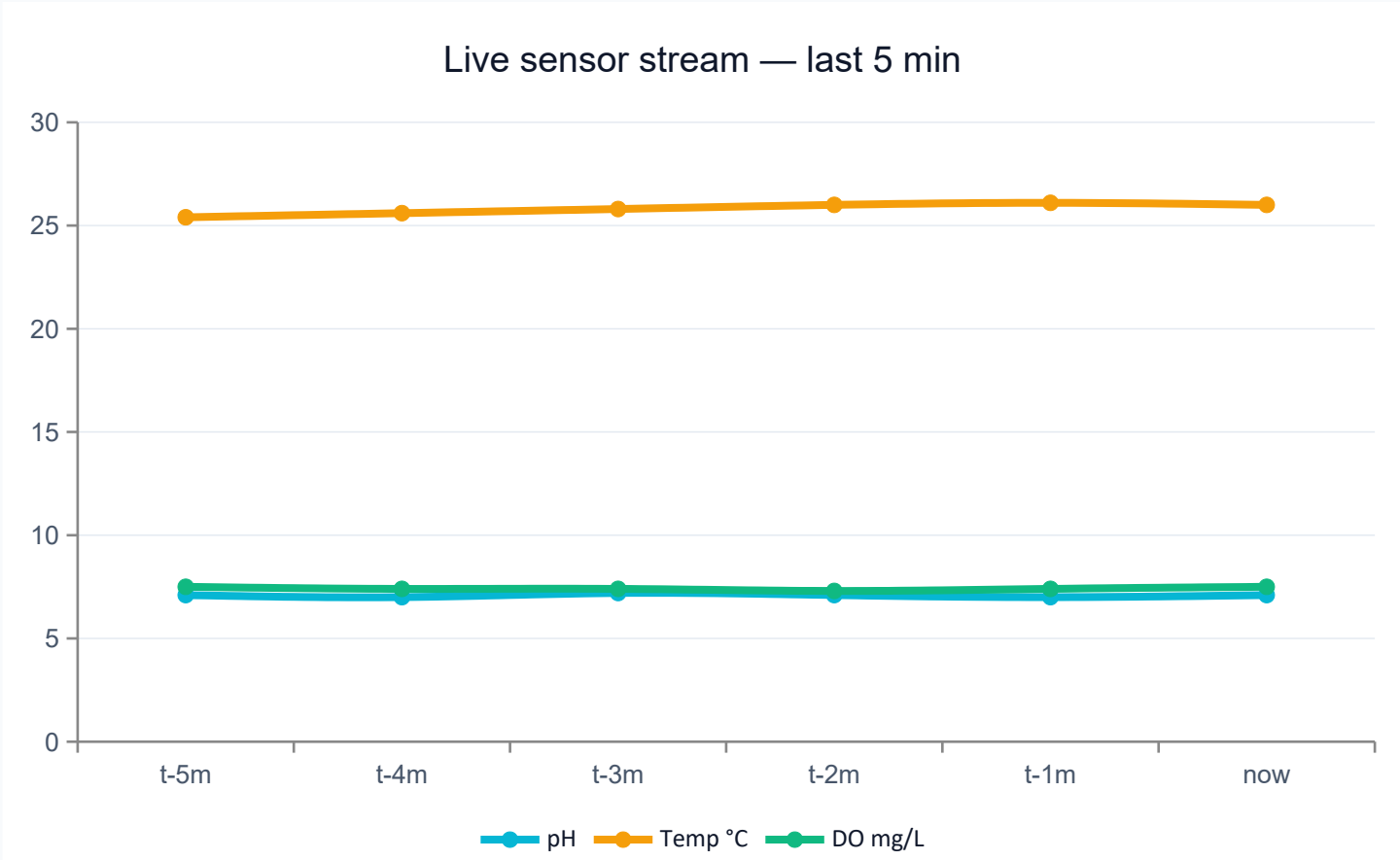
convlstm_vae.pth

Variational autoencoder over 5-frame sequences. Learns 'normal' fish swimming patterns. High reconstruction loss → behavior anomaly (stress, hypoxia, panic).

5 frames

TEMPORAL WINDOW

From Arduino to your phone in 200 ms.



01	Arduino sample	10 ms
	ADC reading + JSON encode	
02	USB serial → Bridge	30 ms
	9600 baud · packet validate	
03	REST → Backend	20 ms
	POST /serial/reading · TypeORM persist	
04	Socket broadcast	5 ms
	sensor:update event to all clients	
05	Client UI render	16 ms
	React state update · 60fps animation	

Cross-platform native UI — built once, runs everywhere.



Live Sensors

Real-time pH, temp, DO gauges with smooth animations.



Tank Controls

Manual feed trigger, pump and LED toggles, schedule editor.



Veronica Chat

Voice + text interface to the AI agent with session history.



Fish Health

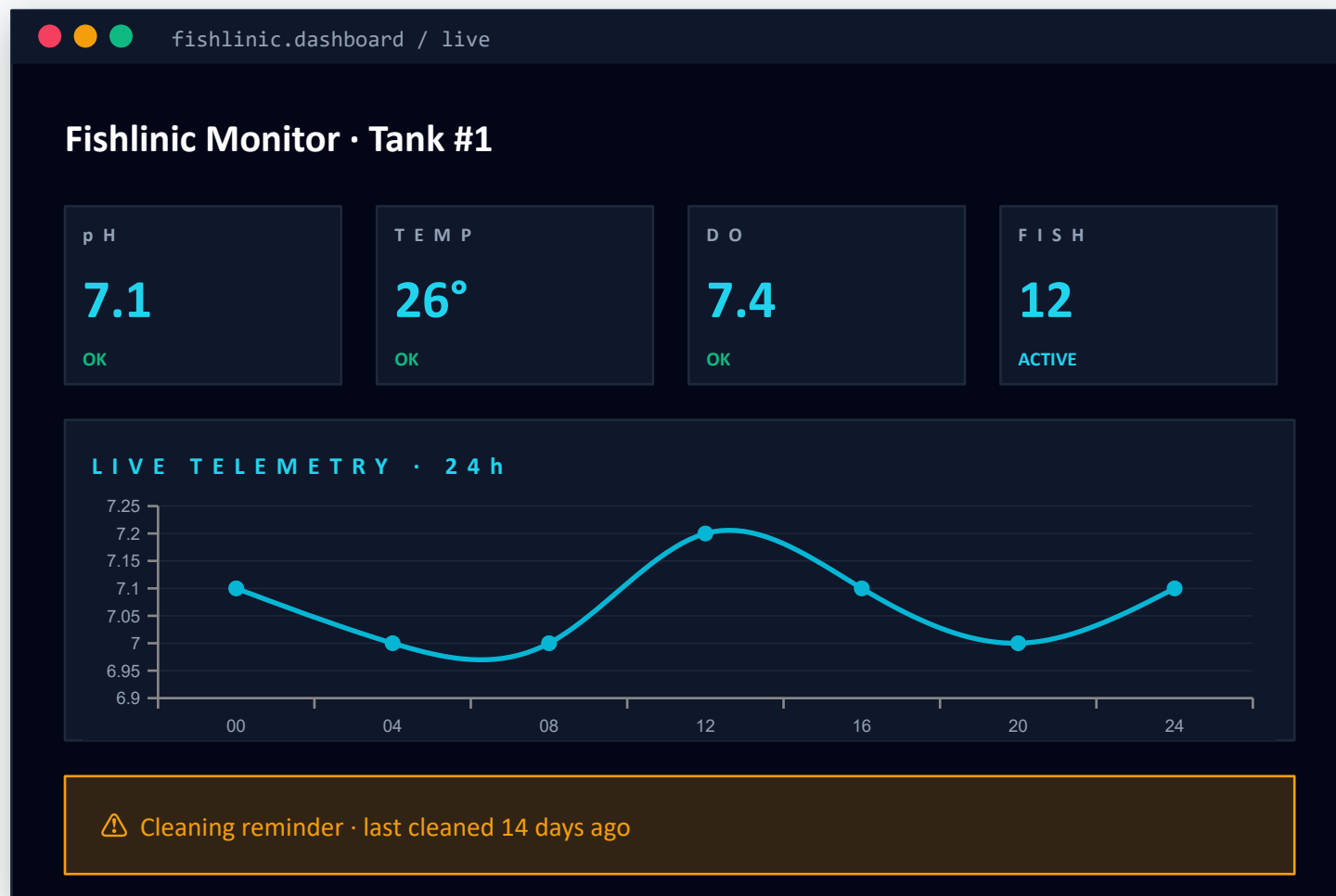
Live disease detection feed with confidence scores and trend.



Push Alerts

Critical thresholds trigger real device push notifications.

Mission control for your aquaculture operation.



Live telemetry charts



One-click hardware controls



Camera feed with detections



Embedded Veronica chat



NextAuth — Google + email



Team profiles + roles

The non-obvious decisions that made it work.



60-second cron tick

Single timer drives all scheduled work — no fragile multiple cron jobs. Feeders, lights, cleaning, alerts all on one beat.



Mock mode everywhere

Backend + bridge fall back to simulated data when hardware is offline. Demos work without an Arduino plugged in.



Bidirectional sockets

Server emits telemetry; clients emit commands. Single Socket.IO gateway handles both directions with type-safe events.



Tool-calling agent

Veronica uses Ollama function-calling to read sensors and act. Agent loop with iteration limit + confirmation gating prevents runaway.



Atomic Design

Atoms → molecules → organisms → screens. Mobile and dashboard share the same component vocabulary across teams.



Shared TypeScript

All services + apps import types from shared/types. One change to a sensor type ripples to mobile, dashboard, backend instantly.

BY THE NUMBERS

Six months of work, shipped.

50K+

Lines of TypeScript

Backend, mobile, dashboard, shared

9

Backend modules

sensors · alerts · actuators · vision · voice · fish · cron · gateway · management

5

AI models

YOLOv11×2, ConvLSTM-VAE, RF, Qwen2.5

3

Client surfaces

Mobile (iOS/Android/Web), Dashboard, Voice

200ms

Sensor → UI latency

Arduino → Bridge → API → Socket → Phone

100%

Offline-capable

Local LLM + mock mode + local DB

How it all comes together — one minute, end to end.



Tank state

Open dashboard. Live pH 7.1, temp 26°C, 12 fish counted by YOLO.

Manual feed

Tap 'Feed' on mobile. Relay clicks. Camera shows fish swarming.

Disease detected

YOLO flags possible fin rot — alert pops up on mobile + dashboard.

Ask Veronica

Voice query: 'how are my fish today?' She reads sensors + diagnoses live.

Autonomous action

Switch to auto mode — Veronica sees high temp, turns on pump unprompted.

THE TEAM

Five engineers. Five disciplines. One product.



Ismail

Lead Architect

Backend (NestJS), Mobile (Expo), Veronica agent, system integration



Maral

Database Specialist

TypeORM schemas, migrations, Supabase production wiring



Hamidullah

Frontend Engineer

Dashboard, atomic UI system, mobile styling polish



Firdavs

AI Engineer

YOLOv11 disease + count, ConvLSTM-VAE, Random Forest pipeline



Sarvar

Hardware Engineer

Arduino firmware (main + secondary), USB serial protocol, relays

From capstone to product.

PHASE 1

Production Hardening

Q3 2026

- Multi-tank scaling (1 backend, N tanks)
- TimescaleDB for sensor history
- AWS / GCP deployment with Terraform
- OAuth role-based access (owner/staff)

PHASE 2

Smarter AI

Q4 2026

- Quantize Veronica to 4-bit (Q4_K_M)
- Fine-tune on aquarium-specific corpus
- Vision-language model for camera Q&A
- Predictive maintenance from sensor trends

PHASE 3

Commercial Pilot

Q1 2027

- Partnership with local fish farms
- Subscription tier with cloud sync
- Industrial-grade sensor packages
- Mobile-first marketplace for replacements

Thank you.

Built with *obsession* for the fish. 

F I S H L I N I C

Smart Aquaculture System · Sejong University Capstone 2026



github.com/Ismoiljon1101/capstone_aquarium_sejong

Questions?